

Technology Stack

Date	15 March 2026
Team ID	
Project Name	Smart Lender: AI/ML-Based Loan Approval Prediction System

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js, HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive, user-friendly interface with reusable components and mobile compatibility.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (Flask)	Lightweight framework for building REST APIs and integrating AI/ML models efficiently.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MySQL	Secure and reliable relational database for storing customer details, loan applications, transaction history, and loan records.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	AWS (EC2 & RDS)	Offers scalable, secure, and cost-effective cloud hosting with managed database services.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub, GitHub Actions	Enables version control, team collaboration, automated testing, and continuous deployment.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Credit Bureau API, Aadhaar/e-KYC API, Twilio SMS API, Email API (SMTP), Scikit-learn	Credit Bureau API retrieves credit scores, Aadhaar/e-KYC verifies identity, Twilio sends OTPs and notifications, Email API sends alerts, and Scikit-learn powers AI-based loan eligibility prediction.

